ModelIt!

Teacher's Guide

Wave Properties: Amplitude, Frequency, and Speed

How Energy and Medium Shape Every Wave in the Universe

Quick Reference

Grade Level	Grade 08L2
NGSS Standard	MS-PS4-1, MS-PS4-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-PS4-1, MS-PS4-2

Use mathematical representations to describe a simple model for waves that includes how the amplitude of a wave is related to the energy in a wave. Develop and use a model to describe that waves are reflected, absorbed, or transmitted through various materials.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model relating wave source energy and medium type to amplitude, frequency, wavelength, and wave speed.
Disciplinary Core Idea	**PS4.A Wave Properties** — A simple wave has a repeating pattern with a specific wavelength, frequency, and amplitude. The relationship between these properties is described by the wave equation.
Crosscutting Concept	**Patterns** — Students identify the mathematical pattern of the inverse relationship between frequency and wavelength, and the consistent pattern of speed = frequency x wavelength across all media.

CAST Testing Alignment

MULTIPLE CHOICE:

A sound wave with a frequency of 440 Hz travels through air at 343 m/s. When the same sound enters water, its speed increases to 1,480 m/s. What happens to the frequency and wavelength of the wave?

CONSTRUCTED RESPONSE:

Using evidence from your model, explain the relationship between amplitude, frequency, and wavelength. Include how changing the wave source energy affects these properties and what happens when a wave moves from one medium to another.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Waves are one of the fundamental ways energy travels through the universe. Sound, light, ocean waves, earthquakes, and radio signals are all waves. Understanding wave properties — amplitude, frequency, wavelength, and speed — reveals the mathematical relationships that govern all wave behavior.

Wave Anatomy

Every wave has four measurable properties: amplitude (height from rest position), wavelength (distance between peaks), frequency (cycles per second, measured in Hz), and speed (distance traveled per second). Amplitude determines how much energy the wave carries. Frequency determines the wave's character — pitch for sound, color for light. Wavelength and frequency are inversely related.

The Universal Wave Equation

The relationship v = f x lambda (speed = frequency x wavelength) is universal — it applies to all waves in all media. If you know any two of these three quantities, you can calculate the third. This equation is one of the most powerful and widely used relationships in all of physics.

Waves and Media

Wave speed depends on the medium. Sound travels at 343 m/s in air, 1,480 m/s in water, and 5,960 m/s in steel. When a wave enters a new medium, its speed changes but its frequency stays the same (because the source is still vibrating at the same rate). Wavelength must adjust to maintain the equation v = f x lambda. Faster speed with same frequency means longer wavelength.

Energy and Amplitude

A wave's energy is proportional to the square of its amplitude. Doubling the amplitude quadruples the energy. This is why a small increase in earthquake magnitude (which relates to wave amplitude) represents a massive increase in energy. The Richter scale is logarithmic — each whole number increase represents about 31.6 times more energy.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify wave source energy, medium type, amplitude, frequency, and wavelength as system components.

E **Establish Relationships**
Students determine that source energy positively affects amplitude and frequency, medium type affects wave speed, and frequency and wavelength have an inverse relationship.

V **Visualize & Model**
Students build the wave property model showing how inputs affect wave characteristics and demonstrating the wave equation.

E **Evaluate Outcomes**
Students compare quiet vs. loud waves and waves in different media to verify the wave equation holds in all scenarios.

R **Revise & Extend**
Students add energy carried or reflection percentage to model more complex wave interactions.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Every sound you hear, every color you see, every earthquake — they're all waves. Let's decode them."*

Watch For

Play a low-pitch and high-pitch tone. Ask: What's different about these two sounds?

Slide 2: Learning Objectives

3 min

Say: *"Today we model the math behind every wave in the universe."*

Watch For

Pre-teach amplitude (height), frequency (speed of vibration), wavelength (distance between peaks).

Slide 3: Big Question

3 min

Say: *"Why does a whisper feel different from a shout? And why does sound change underwater?"*

Watch For

Quick demo: speak normally, then whisper, then shout. What changed physically?

Slide 4: LEVER Framework

2 min

Say: *"We'll model how energy and medium shape wave behavior."*

Watch For

Preview the universal wave equation as the 'punchline' of the lesson.

Slide 5: Activity 1: Components

8 min

Say: *"What can we control? Energy and medium. What responds? Amplitude, frequency, wavelength."*

Watch For

Sort components. Introduce the idea that some properties are mathematically linked.

Slide 6: Activity 2: Connections

8 min

Say: *"Here's the key: frequency and wavelength are inversely related. When one goes up, the other goes down."*

Watch For

Map relationships. Highlight the inverse connection — this is the most important pattern.

Slide 7: Activity 3: Simulation

10 min

Say: *"Watch what happens when we change the medium. Speed changes, frequency stays. What adjusts?"*

Watch For

Students predict before running. Focus on wavelength adjusting to maintain v = f x lambda.

Slide 8: Discoveries

5 min

Say: *"Speed equals frequency times wavelength. This works for sound, light, ocean waves — everything."*

Watch For

Calculate v = f x lambda using the simulation data. Verify it works in every scenario.

Slide 9: STEM Challenge

12 min

Say: *"Design a sonar system for a submarine. What frequency and energy do you need? The math will tell you."*

Watch For

Teams design sonar specifications using wave equation calculations. Present and defend.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Wave Source Energy, Medium Type
- **INTERNAL:** Amplitude, Frequency, Wavelength

Relationships

- **Wave Source Energy to Amplitude:** POSITIVE (+) — More energy from the source creates larger wave amplitude — louder sounds, brighter light, taller ocean waves.
- **Wave Source Energy to Frequency:** POSITIVE (+) — More energy can drive higher-frequency oscillations — though frequency is primarily set by the source mechanism.
- **Medium Type to Wave Speed:** POSITIVE/NEGATIVE (+/-) — Denser media generally increase wave speed for mechanical waves (sound in water is faster than in air) because particles are closer together and transmit vibrations more efficiently.
- **Frequency to Wavelength:** NEGATIVE (-) — Frequency and wavelength are inversely related. Higher frequency waves have shorter wavelengths. This is the core mathematical relationship: if v is constant, then lambda = v/f.
- **Amplitude to Energy Carried:** POSITIVE (+) — Larger amplitude waves carry more energy. Energy is proportional to the square of amplitude — doubling amplitude quadruples the energy carried.

Key Discoveries

- More source energy increases amplitude (louder/brighter) and can increase frequency (higher pitch)
- Frequency and wavelength have an inverse relationship — higher frequency means shorter wavelength
- When a wave enters a new medium, its speed changes but its frequency stays the same
- The wavelength adjusts to match the new speed while maintaining the original frequency
- The wave equation v = f x lambda (speed = frequency x wavelength) holds true in every medium

Multiple Choice Post-Assessment Answer Key

1. Your model shows that when a sound wave moves from air (343 m/s) to water (1,480 m/s), wave speed increases but frequency stays the same. Using the wave equation v = f x wavelength, what must happen to wavelength?

Correct: B) Wavelength increases proportionally because speed increased while frequency remained constant

Correct! From v = f x wavelength, if speed (v) increases and frequency (f) stays constant, wavelength must increase to maintain the equation. The wave stretches out in the denser medium while oscillating at the same rate.

2. In the model, increasing Wave Source Energy from 20% to 90% increased Amplitude dramatically. A student asks: does doubling the amplitude double the energy? What does the model reveal?

Correct: B) No, energy is proportional to the square of amplitude. Doubling amplitude quadruples the energy carried by the wave.

Correct! Energy is proportional to amplitude squared. This is why a small increase in earthquake magnitude (related to wave amplitude) represents a huge increase in destructive energy. The relationship is exponential, not linear.

3. Your model shows a NEGATIVE relationship between Frequency and Wavelength. Which real-world example best illustrates this inverse relationship?

Correct: A) Radio waves have low frequency and long wavelength, while gamma rays have high frequency and short wavelength

Correct! The electromagnetic spectrum perfectly demonstrates the inverse relationship. Radio waves oscillate slowly (low frequency) with long wavelengths, while gamma rays oscillate rapidly (high frequency) with very short wavelengths. In any given medium, speed = frequency x wavelength.

4. A student observes that frequency stays the same when a wave enters a new medium, even though speed and wavelength change. Why does frequency remain constant?

Correct: B) Because frequency is determined by the source, which continues vibrating at the same rate regardless of what medium the wave enters

Correct! Frequency is set by the source. A tuning fork vibrating at 440 Hz pushes air molecules 440 times per second. When those vibrations reach water, the water molecules also vibrate 440 times per second because they are being driven by the same source. The medium changes speed and wavelength, not frequency.

5. Based on your model, a marine scientist needs sonar to detect small objects on the ocean floor. High-frequency sonar provides fine detail but has short range, while low-frequency sonar travels far but provides less detail. Which model evidence supports choosing high-frequency sonar for this task?

Correct: B) High-frequency waves have shorter wavelengths, which can resolve smaller features because the wavelength is closer to the size of the objects being detected

Correct! The model shows the inverse relationship between frequency and wavelength. Higher frequency means shorter wavelength, and waves can only resolve objects roughly the size of their wavelength or larger. For detecting small objects, short wavelengths (high frequency) provide the necessary resolution.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Louder sounds have higher frequency"	Loudness and pitch are independent properties. Loudness relates to AMPLITUDE (wave height), while pitch relates to FREQUENCY (vibrations per second). You can whisper a high note or shout a low note — amplitude and frequency are separate.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Sound travels at the same speed everywhere"	Sound speed depends entirely on the medium. In air, sound travels at about 343 m/s. In water, it's 1,480 m/s. In steel, 5,960 m/s. The density and elasticity of the medium determine how fast vibrations propagate.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Waves carry matter from one place to another"	Waves carry ENERGY, not matter. The particles in the medium oscillate back and forth but stay in place — they don't travel with the wave. A cork floating on ocean waves bobs up and down but doesn't move toward shore. The wave pattern moves, not the water.